BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: 25-26J-138

Project Title: Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

1. Introduction
   1.1. Background:

Mental health challenges among university students have become a growing global concern, with stress, anxiety, and cognitive overload increasingly affecting academic performance, emotional stability, and overall well-being. Undergraduate students frequently experience pressure arising from academic workload, examinations, social expectations, financial responsibilities, and uncertainty regarding future career prospects. When left unmanaged, prolonged stress can negatively impact learning ability, decision-making, and long-term mental health outcomes.

Traditional mental health support mechanisms such as counseling services and therapy sessions play an important role in addressing these challenges. However, such services often face limitations related to accessibility, availability, cost, stigma, and delayed intervention. As a result, many students do not receive timely support, particularly during early stages of stress escalation. This gap highlights the need for scalable, accessible, and technology-driven solutions that can provide continuous support.

Advancements in Artificial Intelligence (AI) and data analytics have enabled the development of intelligent conversational agents, commonly known as chatbots, which are increasingly used in mental health support systems. These AI-powered chatbots can engage users in real-time conversations, provide emotional support, and offer coping strategies. Despite their growing popularity, most existing chatbots rely primarily on black-box machine learning models or pattern-based Natural Language Processing (NLP) techniques. Such approaches often lack transparency, contextual reasoning, and adaptability to individual users over time.

Recent research emphasizes the importance of explainable and human-like reasoning in mental health applications. Neuro-symbolic AI, which integrates neural networks with symbolic reasoning and domain knowledge, offers a promising direction to address these limitations. By combining data-driven learning with psychological logic and behavioral understanding, neuro-symbolic systems can provide explainable, context-aware, and personalized support. When enhanced with cognitive and behavioral indicators such as linguistic patterns, interaction dynamics, and temporal emotional changes, these systems can offer more accurate stress detection and intervention strategies.

This research focuses on leveraging data analysis techniques to support the development of an adaptive, explainable, and intelligent chatbot framework for stress regulation among university students.

   1.2. Research Problem:

- Increasing stress and cognitive overload among university students negatively affect academic performance and overall well-being.
- Existing traditional and digital stress-management solutions are not scalable and have limited accessibility for all students.
- Current approaches do not sufficiently analyze emotional, behavioral, and interaction data to identify student stress patterns accurately.
- Lack of data-driven and personalized stress analysis that adapts to individual student differences over time.
- Insufficient analytical support for early detection and timely stress regulation interventions.

1.3. Objectives:

It Number	Objective	Objective number
IT22003928	Enable transparent and explainable stress interpretation using neuro-symbolic emotion reasoning.	1
IT22332912	Detect real-time cognitive overload using passive typing behavior and linguistic patterns.	2
IT22886118	Support long-term emotional adaptation through continual learning without losing historical patterns.	3
IT22628404	Determine the optimal timing and appropriate multi-modal format for delivering stress-regulation interventions.	4

2. Data Exploration

2.1. Data Collection

The data for this study were collected using a combination of secondary and primary sources to ensure a comprehensive understanding of the research topic.

1. Secondary Data:

We leveraged open-source datasets available on Kaggle that are relevant to the study variables, such as stress levels, demographic factors, academic performance, and lifestyle indicators among university students. These datasets provided large-scale quantitative data that allowed for preliminary analysis and identification of trends. In addition to Kaggle, data were extracted from published research papers, reports, and institutional studies on stress among undergraduates, which provided insights into

validated measurement scales, common stressors, and previously observed correlations. These sources were critically reviewed to complement the primary data and to triangulate findings.

2. Primary Data (Survey):
To capture specific and current information, a structured survey was conducted among undergraduates of the Sri Lanka Institute of Information of Technology, Informatics Institute of Technology and University of Colombo. The survey was created as a survey website included both closed-ended questions (using Likert scales for stress levels, frequency of stressors, and coping mechanisms) and open-ended questions (allowing respondents to elaborate on personal experiences and challenges). The survey was distributed electronically via email and social media platforms and done in person as well, ensuring voluntary participation and anonymity to encourage honest responses.

3. Ethical Considerations:
All participants were informed about the purpose of the study, and consent was obtained prior to data collection. The data collected were kept confidential and used exclusively for research purposes.

By combining secondary datasets and a primary survey, the study ensures robustness and reliability of the data, enabling both quantitative analysis and qualitative insights into stress patterns among university undergraduates.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
StressID Dataset[1]	A multimodal dataset designed for stress identification containing synchronized physiological, video, and audio data from participants performing stress-inducing and relaxation tasks.	NeurIPS 2023 StressID dataset (provided via research access on project site)	~39 hours of annotated data across modalities (~5.29 GB of raw files)	• **Participants:** 65 individuals performing 11 tasks. • **Physiological signals:** ECG (electrocardiography), EDA (electrodermal activity), respiration. • **Video recordings:** facial expression videos (RGB). • **Audio recordings:** speech and task-related audio in WAV. • **Self-assessment labels:** perceived stress, relaxation, arousal, valence for each task.

				• **Derived labels for ML:** binary and three-class stress labels (e.g., stressed vs. not stressed).
ConvCounsel Dataset [2]	A conversational dataset for student mental health counseling, containing real dialogues between students and counselors designed to support the development of AI-powered counseling systems. The dataset emphasizes active listening strategies used in counseling conversations and includes both speech and text components for multimodal analysis and model training.	ConvCounsel: A Conversational Dataset for Student Counseling, IEEE O-COCOSDA 2024 proceedings (available via research access)	Contains conversational data with speech and text across a large number of counseling dialogues ~9GB (e.g., thousands of conversational turns) designed for training and evaluating mental health dialogue systems. Estimated dataset size varies by version and modality (text and audio). (AI Models)	• **Speech recordings:** counselor–student verbal exchanges (audio) • **Text transcripts:** annotated conversation text from counseling sessions • **Dialogue context:** multi-turn counseling exchanges with counseling strategy annotations (e.g., active listening) (AI Models) • **Mental health themes:** topics include stress, anxiety, academic/challenge concerns, emotional support dialogue (AI Models)
Behavioral Interaction Dataset	Passive behavioral interaction logs collected during task execution	Custom-built research website	Session-based dataset collected from multiple users (scalable)	• Mouse movement speed and trajectory • Mouse click frequency and dwell time • Keystroke latency and inter-key delay • Task completion time • Session timestamps • Derived behavioral metrics (hesitation, variability)

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

(Explain how well each dataset aligns with your research problem and objectives)

	1	2	3	4
Data source 1	x		x	x
Data source 2	x		x	x
Data source 3		x		

3. Methodology

3.1. Data Preprocessing:

(Mention data transformation techniques done in each dataset for each objective.)

Ex:

Data Cleaning, Data Normalization, Data Standardization, Data Encoding (e.g., One-Hot Encoding, Label Encoding), Handling Missing Data (e.g., Imputation or Removal), Data Aggregation, Feature Engineering, Outlier Detection and Handling, Data Scaling, Data Discretization, Dimensionality Reduction (e.g., PCA), Date/Time Transformation, Data Integration (Merging or Joining), Data Mapping, Data Type Conversion.

Objective 1

	Transformation technique							
Dataset	**Data Cleaning**	**Data Normalization / Standardization**	**Data Encoding**	**Handling Missing Data**	**Data Aggregation**	**Feature Engineering**	**Outlier Detection / Handling**	**Data Sc**
Data source 1	- Removed corrupted audio files - Filtered background noise using band-pass and notch filters - Corrected misaligned or incomplete transcripts - Removed silent or irrelevant segments in audio	- Normalized audio amplitude to [-1, 1] - Standardized feature ranges for physiological consistency across participants	- Label Encoding for stress classes (binary: stressed/not stressed; multi-class: low/medium/high) - Tokenized text - Converted words into numeric sequences for embeddings	- Interpolated missing audio signal segments - Manually verified missing text transcriptions and corrected	N/A (features used per instance)	- Audio: MFCC, Chroma, Spectral Contrast, Zero-Crossing Rate, pitch, energy - Text: Sentiment scores, keyword frequency, dialogue acts, TF-IDF vectors, Word2Vec embeddings	- Detected extreme audio values using z-score thresholds (e.g., amplitude spikes) - Removed text outliers such as corrupted characters	- Standa numeric features zero me unit vari for ML m
Data Source 2	- Removed transcription errors and	- Normalized audio features (MFCC, pitch,	- One-Hot Encoding for dialogue acts	- Removed incomplete audio or text	N/A	- Audio: prosodic features	- Removed extreme audio	- Standar audio

	irrelevant symbols - Removed non-verbal noises from audio (e.g., coughs, laughter) - Removed duplicate or repeated turns	energy) across all sessions to reduce participant-specific variations	and emotional state labels - Label Encoding for target stress/emotion classes	segments - Ignored sessions with >20% missing data		(pitch, intensity, energy), spectral features, MFCCs - Text: sentiment analysis, dialogue acts, keyword frequency, n-grams	feature values (pitch, amplitude) - Checked transcripts for unusual token lengths	numeri feature ML - Standar text embedd values
Data Source 3	- Removed incomplete or corrupted session logs - Filtered noisy mouse movement records caused by browser glitches - Removed idle periods and irrelevant interaction events (e.g., tab switches, inactive states) - Corrected inconsistent timestamp formats	- Normalized time-based features (task duration, dwell time) to a common range - Standardized interaction frequency metrics to ensure consistency across users	- Encoded categorical task types using label encoding - Converted interaction states (active/inactive) into binary values - Mapped task difficulty levels to numeric values	- Missing interaction values were imputed using mean/median based on session context - Sessions with excessive missing data were removed	- Aggregated raw interaction-level data into session-level features (mean mouse speed, total clicks, average keystroke delay)	- Mouse features: speed, acceleration, trajectory deviation, hesitation rate - Keyboard features: keystroke latency, inter-key delay, typing variability - Temporal features: task duration, pause frequency, idle time ratio	- Detected extreme behavioral values using IQR and Z-score thresholds (e.g., abnormal mouse spikes) - Removed outlier sessions caused by technical errors	- Standar numeri feature zero me and uni varianc ensure compat with machin learnin models

Objective 2

	Transformation technique							
Data source 1								

Objective 3

	Transformation technique							
Data source 1								

Objective 4

	Transformation technique							
Data source 1								

3.2. Scalability

(Assess whether each dataset size is sufficient and scalable for analysis.)

3.3. Feature extraction

(Mention steps followed to evaluate the key features and attributes in datasets for each objective)

4. Modelling and Results

The results section should present the key findings derived from analyzing the data. It should answer the research questions or hypotheses posed at the beginning of the study, based on the methods and techniques applied during the analysis.

4.1. Key Insights:

(Summarize key findings from each dataset exploration and analysis. Provide an in-depth exploration of the data using visualizations such as charts, graphs, and plots. Describe patterns, trends, and correlations between variables)

4.2. Challenges Faced During Data Analysis:

(Discuss any challenges encountered in working with the datasets, e.g., missing data, outliers, or computational limits.)

5. References

(List all academic references, datasets, and other sources used in the report, following the chosen IEEE citation style.)